

The Physiology of Response: When an Interaction Is Relevant to an Agent, and What a Relevant Response Costs

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Abstract

An agent embedded in a busy world is addressed constantly, and most of what reaches it does not matter to it. We ask what it is for an incoming *interaction* to be *relevant* to an agent, and what a relevant response *costs*, and we answer with mathematics a reader can follow without any other work. We take an agent to have two standing quantities, which we define from scratch: a *purpose*, the attractor of its own motivation, and a *coherence*, the internal consistency it must not break. A candidate response to an interaction is admissible only if it is *two-factor relevant*: it must (a) advance the agent toward its purpose and (b) preserve the agent’s coherence. From four axioms—bounded attention, a positive floor on the cost of any response, purpose-directed drive, and coherence as a closed consistency condition—we prove: **(T1)** relevance is the conjunction of a purpose gain and a coherence margin, and neither alone suffices; **(T2)** an agent has a bounded attention budget, so it can commit to at most finitely many responses per unit time and must decline the rest—inattention is forced, not a defect; **(T3)** the agent’s optimal policy is a threshold rule: respond exactly to interactions whose purpose gain exceeds a price set by the attention budget, and ignore the rest, which reproduces the lived phenomenology of an agent that is present but preoccupied; **(T4)** coherence is testable locally and in advance—an incoherent response is detectable from the signs of its internal effects alone, before it is committed, so an agent can refuse a purpose-advancing re-

sponse that would break it; **(T5)** response is floored and irreversible: every committed response deposits a positive, unrepealable change in the agent, so an agent cannot un-receive what it has responded to, and successive encounters are never with the same agent; **(T6)** construction and commitment are mutually exclusive at each instant—an agent forming new distinctions cannot simultaneously commit to an action—so responsiveness and reflection alternate, which is a second, independent source of the appearance of distraction. We give the admissible-response construction, the threshold policy, the coherence test, and a numerical study on explicit agents. The reading of the formal apparatus as attention, preoccupation, and the felt autonomy of a character is marked as such; no theorem depends on it.

Keywords: relevance; attention budget; threshold policy; coherence; holonomy; irreversibility; construction–commitment alternation; non-player character; agent response.

1 Introduction

1.1 The question

An agent in a shared world receives far more than it can act on. A passer speaks to it; an event occurs nearby; an object enters its field. Most of this is nothing to the agent—not because the agent fails to perceive it, but because responding to it would not serve what the agent is doing and is not worth the agent’s limited capacity. The mark of a convincing embedded agent is precisely this selectivity: it is present,

but it is not at anyone’s disposal. It answers what matters to it and lets the rest pass, sometimes with a token acknowledgement, sometimes with none. This selectivity is the response side of the synthetic-agent problem [14, 17].

This paper asks what makes an incoming interaction *relevant* to an agent, and what a relevant response *costs*. We want a definition sharp enough to compute a policy from—given an interaction, should the agent respond, and if so how—and self-contained: nothing but this paper and standard mathematics.

1.2 The two-factor thesis

We argue that relevance has two factors and needs both. A response to an interaction is worth making only if it *advances the agent toward its purpose* and *preserves the agent’s coherence*. The first factor is self-interest: a response that does not move the agent toward what it is trying to do is, to the agent, idle. The second is integrity: a response that would advance the purpose but break the agent’s internal consistency—make it act against itself—is inadmissible. Relevance is the conjunction. An interaction to which no response passes both tests is irrelevant to the agent, and the correct behaviour is to decline it.

From this, together with the plain fact that the agent’s capacity to respond is bounded, the phenomenology of an embedded character follows: selective attention, declined interactions, token acknowledgements, and the alternation between engaging and withdrawing.

1.3 Relation to the companion account of purpose

This paper is self-contained; it defines every quantity it uses. It does, however, take from a companion study the picture of an agent as having a standing *identity* and a *purpose* that attracts its conduct. We restate exactly what we need—purpose as the attractor of a drive, and coherence as a closed consistency condition—as Section 2 axioms and definitions, proved or posited here, so that the present paper stands alone. A reader who has not seen the companion account loses nothing.

1.4 Organisation

Section 2 fixes the agent’s standing quantities—purpose and coherence—and the interaction and response. Section 3 proves the two-factor relevance theorem (T1). Section 4 proves the bounded-attention and threshold-policy results (T2, T3). Section 5 proves that coherence is testable in advance (T4). Section 6 proves response is floored and irreversible (T5). Section 7 proves construction–commitment alternation (T6). Section 8 reports a numerical study. Section 9 discusses scope.

2 The agent’s standing quantities, the interaction, the response

We fix the objects. The agent has a space of dispositions, a purpose it is drawn to, and a coherence it must keep. An interaction arrives; a response is a change the agent may make to itself in reply.

2.1 Dispositions, purpose, drive

Definition 2.1 (Disposition space). The *disposition space* $\mathcal{B}$ of an agent is a compact convex subset of $\mathbb{R}^n$; a point $x \in \mathcal{B}$ is the agent’s current disposition (its tendencies to act). Distance on $\mathcal{B}$ is Euclidean, $d(x, y) = \|x - y\|$.

Axiom 2.2 (Purpose-directed drive). The agent carries a *drive potential* $\Phi : \mathcal{B} \rightarrow \mathbb{R}$, continuously differentiable and λ -strongly convex for some $\lambda > 0$ (standard convex analysis [13, 2, 12]): $\Phi(y) \geq \Phi(x) + \nabla\Phi(x)^\top(y - x) + \frac{\lambda}{2}\|y - x\|^2$ for all x, y . Its behaviour, absent interaction, follows the descent $\dot{x} = -\nabla\Phi(x)$ [9, 7].

Definition 2.3 (Purpose). The *purpose* of the agent is the unique minimiser $x^* = \arg \min_{x \in \mathcal{B}} \Phi(x)$; it is the attractor of the drive. For a disposition x , the *residual to purpose* is $D(x) = \Phi(x) - \Phi(x^*) \geq 0$, the drive-height still to be descended.

Lemma 2.4 (Purpose is well-defined and attracting). *Under Axiom 2.2, x^* exists and is unique, and along the drive $\|x(t) - x^*\| \leq e^{-\lambda t}\|x(0) - x^*\|$; equivalently $D(x(t)) \leq e^{-2\lambda t}D(x(0))$.*

Proof. A continuous function on a compact set attains its minimum; strong convexity makes the minimiser unique. For attraction, let $V = \frac{1}{2}\|x - x^*\|^2$; then $\dot{V} = -(x - x^*)^\top \nabla \Phi(x) \leq -\lambda\|x - x^*\|^2 = -2\lambda V$ using strong-convexity monotonicity and $\nabla \Phi(x^*) = 0$, so $V(t) \leq e^{-2\lambda t}V(0)$ by Grönwall. The D -form follows since D is comparable to $\frac{\lambda}{2}\|x - x^*\|^2$ from below and to $\frac{L}{2}\|x - x^*\|^2$ from above for the gradient’s Lipschitz constant L . $\square$

2.2 Coherence

An agent is coherent when its dispositions do not contradict one another: when the ways it is disposed to act, taken around any internal loop of dependence, agree. We formalise this as the vanishing of a *holonomy*—the net drift accumulated on going around a cycle—which is the standard way to say “all routes agree.”

Before defining the consistency graph we must say what *gives* the agent a transport structure at all: the graph must be grounded in the agent’s internal constitution, not introduced fully formed.

Axiom 2.5 (Disposition components carry transports). The agent’s disposition components stand in relations of internal dependence: shifting one component induces, through the agent’s own internal structure, a determinate shift in each dependent component. Each such directed dependence carries a *transport*—the shift induced per unit of the driving component. Formally, the agent’s internal structure supplies, for each ordered pair of dependent components (i, j) , a transport $\theta_{ij} \in \mathbb{R}^d$; the consistency graph H of Definition 2.7 is the formalisation of this transport structure.

Remark 2.6 (The consistency graph is the self-graph carrying its costs as transports). The transport structure of Axiom 2.5 is not a new object bolted onto the agent but the companion account’s self-graph seen dynamically. In that account the agent is a finite weighted graph of internal distinctions whose edges carry the *cost* of holding two parts apart; a shift in one part is transmitted to an adjacent part through exactly that separation, and the induced shift per unit drive is the transport θ_{ij} . The consistency graph H shares the self-graph’s vertices (disposition components) and its edges (internal separa-

rations), and reads each separation cost as the magnitude of a directed transport. Holonomy—the failure of transports to sum to zero around a cycle—is then the dynamical signature of the same internal boundary costs the companion account derives, so H arises from the agent’s constitution rather than alongside it. This is what grounds the pre-commitment coherence test (Theorem 5.1) and the refusal corollary (Corollary 5.2): both read structure the agent already has.

Definition 2.7 (Consistency graph; holonomy; coherence). The agent’s *consistency graph* H has a vertex for each disposition component and a directed edge $i \rightarrow j$ carrying a *transport* $\theta_{ij} \in \mathbb{R}^d$ recording the shift the agent’s structure induces in j per unit of i . The *holonomy* of a directed cycle $c = (i_0, i_1, \dots, i_0)$ is $\eta(c) = \sum_\ell \theta_{i_\ell i_{\ell+1}}$. The agent is *coherent* if $\eta(c) = 0$ for every cycle c ; the *coherence margin* of a disposition x is $\kappa(x) = -\max_c \|\eta_x(c)\|$, the negative of the worst cycle disagreement at x (so $\kappa(x) = 0$ at perfect coherence and $\kappa(x) < 0$ when some cycle fails to close).

Axiom 2.8 (Coherence is a closed condition). The set of coherent dispositions $\{x \in \mathcal{B} : \kappa(x) = 0\}$ is closed and nonempty, and $\kappa : \mathcal{B} \rightarrow (-\infty, 0]$ is continuous. (An agent has at least one coherent disposition and coherence does not jump.)

Remark 2.9 (Why holonomy is the right notion). “All routes agree” is exactly “every cycle sums to zero.” If two chains of internal dependence lead from a disposition to another and disagree, their difference is a cycle with nonzero holonomy; conversely a nonzero cycle is two routes that disagree. This is the graph- and geometry-theoretic notion of holonomy and cycle space [1, 5, 6, 10]. Coherence as vanishing holonomy is therefore not a special assumption but the general form of internal consistency, and—crucially for Section 5—it is checkable from the *signs and sums of local transports*, without any global solve.

2.3 Interaction and response

Definition 2.10 (Interaction). An *interaction* ι is an external event presented to the agent: formally a map $\iota : \mathcal{B} \rightarrow \mathcal{B}$ that the agent *may* apply to its disposition (the change the

interaction would induce if the agent takes it up), together with a tag identifying it. An interaction is *received* whether or not it is responded to; it is *responded to* only if the agent applies a response to it.

Definition 2.11 (Response; repertoire; cost). A *response* to ι is a map $r : \mathcal{B} \rightarrow \mathcal{B}$ the agent applies to its own disposition in reply, drawn from a finite *repertoire* $\mathcal{R}$. Applying r moves the disposition from x to $r \cdot x$. Each response has a *cost* $c(r) > 0$, the capacity it consumes.

Axiom 2.12 (Response floor). No response is free: there is $\beta > 0$ with $c(r) \geq \beta$ for every $r \in \mathcal{R}$. Even the smallest genuine response—a token acknowledgement—costs at least β .

Remark 2.13 (Why the floor is constitutive of response). The floor is not an independent stipulation but is built into what a response *is*. A zero-cost response deposits nothing and changes nothing—it leaves the agent’s disposition, expended attention, and committed count exactly as they were. But a change that changes nothing is not a response; it is the *absence* of response, indistinguishable from declining. Therefore every genuine response—anything that counts as the agent having replied at all—costs at least some positive $\beta > 0$. The floor is thus the trace that distinguishes responding from doing nothing, and Axiom 2.12 records the least such trace across the repertoire.

Axiom 2.14 (Bounded attention). The agent has a finite *attention budget* $\alpha > 0$ per unit time: the total cost of the responses it commits in any unit interval cannot exceed α . Formally, if R is the set of responses committed in a unit interval, $\sum_{r \in R} c(r) \leq \alpha$. This finiteness is the formal counterpart of bounded rationality [16].

3 Two-factor relevance

We now define when an interaction is relevant and prove that relevance is the conjunction of a purpose gain and a coherence margin, with neither sufficient alone.

Definition 3.1 (Purpose gain). The *purpose gain* of a response r to ι at disposition x is the residual it removes,

$$g(r; x) = D(x) - D(r \cdot x) = \Phi(x) - \Phi(r \cdot x),$$

the drop in drive-height. The response is *purpose-advancing* at x if $g(r; x) > 0$. Reading the residual D as a value-to-go and the gain as its reduction places the policy in the setting of reward and value [18], while the residual itself is an information-like gap in the sense of [15, 4].

Definition 3.2 (Coherence-preserving). A response r is *coherence-preserving* at x if it does not worsen the coherence margin: $\kappa(r \cdot x) \geq \kappa(x)$, i.e. the agent is no less coherent after responding than before. It is *coherence-breaking* if $\kappa(r \cdot x) < \kappa(x)$.

Definition 3.3 (Two-factor relevance). An interaction ι is *relevant* to the agent at disposition x if there exists a response $r \in \mathcal{R}$ that is both purpose-advancing and coherence-preserving at x :

$$\exists r \in \mathcal{R} : g(r; x) > 0 \text{ and } \kappa(r \cdot x) \geq \kappa(x).$$

Such a response is *admissible*. An interaction with no admissible response is *irrelevant* to the agent.

Theorem 3.4 (Relevance is a conjunction; neither factor alone suffices). Under Axioms 2.2 and 2.8:

- (i) *An interaction is relevant iff its admissible set is nonempty, i.e. iff some response clears both the purpose-gain and coherence-margin bars.*
- (ii) *Purpose gain alone does not make an interaction relevant: there exist interactions whose every purpose-advancing response is coherence-breaking, hence inadmissible.*
- (iii) *Coherence preservation alone does not make an interaction relevant: a response that preserves coherence but has $g \leq 0$ is inadmissible.*

Proof. (i) is Definition 3.3 restated: relevance is nonemptiness of the admissible set, and admissibility is the conjunction of the two conditions. (ii) Construct $\mathcal{B} = [0, 1]^2$, a strongly convex Φ with minimiser at a corner, and a repertoire whose only $g > 0$ response carries the disposition across a locus where some cycle’s holonomy jumps from 0 to a positive norm, so κ strictly decreases; then every purpose-advancing response is coherence-breaking, the

admissible set is empty, and the interaction is irrelevant despite offering purpose gain. (Continuity of κ , Axiom 2.8, makes the decrease well-defined.) (iii) If $g(r; x) \leq 0$ then r is not purpose-advancing, so it fails the first conjunct and is inadmissible regardless of its coherence effect. $\square$

Remark 3.5 (The peripheral player, derived). Theorem 3.4 already yields the central phenomenological fact. An interaction from another party (a player addressing the agent) is relevant only if some response to it both advances the agent’s own purpose and keeps the agent coherent. When the player’s address does neither—as is typical when the agent is busy with its own pursuit—the interaction is simply irrelevant, and the agent’s correct behaviour is to not respond, or to respond only with the cheapest coherence-neutral token. The player is peripheral by construction: relevance is measured against the agent’s purpose, not the player’s. No special “distraction” mechanism is needed; peripherality is what relevance-to-self means.

4 Bounded attention and the threshold policy

Relevance says which interactions *could* be worth a response. Bounded attention says the agent cannot take up all of them, and forces a selection. We show the selection is a threshold rule and that declining interactions is not a failure but the optimal use of a finite budget.

Theorem 4.1 (Inattention is forced). *Under Axioms 2.12 and 2.14, in any unit interval the agent commits at most $\lfloor \alpha/\beta \rfloor$ responses. If more than $\lfloor \alpha/\beta \rfloor$ relevant interactions arrive in the interval, the agent must leave at least one relevant interaction without response. Declining interactions is therefore forced by finiteness, not a defect of the agent.*

Proof. Each committed response costs $\geq \beta$ (Axiom 2.12) and the total committed cost is $\leq \alpha$ (Axiom 2.14); hence the number of committed responses N satisfies $N\beta \leq \alpha$, i.e. $N \leq \lfloor \alpha/\beta \rfloor$. If the number of relevant interactions exceeds this bound, not all can be committed, so at

least one relevant interaction goes without response. $\square$

Definition 4.2 (Response policy; value). A *response policy* selects, from the relevant interactions arriving in a unit interval, a subset R to respond to, subject to $\sum_{r \in R} c(r) \leq \alpha$. Its *value* is the total purpose gain secured, $\sum_{r \in R} g(r; x)$ (each interaction answered by its best admissible response).

The responses of Definition 2.11 are drawn from a *finite* repertoire: they are discrete, all-or-nothing, not divisible at will. The value-maximising selection is therefore an integer knapsack, which is NP-hard in general [11, 8, 3]. Two honest positions are available, and we take both: the threshold rule is *exactly* optimal when the repertoire has the additive structure below, and otherwise optimal for the continuous relaxation and within one boundary item of the true integer optimum.

Definition 4.3 (Floor-additive repertoire). A response repertoire is *floor-additive* if every response decomposes into acknowledgement-sized units of cost β with proportional gain: each r has $c(r) = k_r \beta$ for some integer $k_r \geq 1$, and gain accrues linearly across its units, so that committing k_r units of the cheapest floor-cost response secures the same gain density as r . Equivalently, cost is additive and gain is additive over floor-sized units.

Theorem 4.4 (Optimal policy is a threshold on gain density). *Order relevant interactions by gain density $g(r; x)/c(r)$ and commit them in decreasing order until the budget α is exhausted; equivalently, there is a price $p^* \geq 0$ such that the agent responds to an interaction iff its best admissible response has gain density $\geq p^*$, and ignores the rest, with p^* rising as the budget α falls or as relevant arrivals grow. Then:*

- (i) *for a floor-additive repertoire (Definition 4.3) the threshold rule is exactly value-maximising among all integer selections; and*
- (ii) *for a general finite repertoire it is optimal for the continuous relaxation and its value is within one response of the integer optimum—the gap is at most the gain of a single boundary item.*

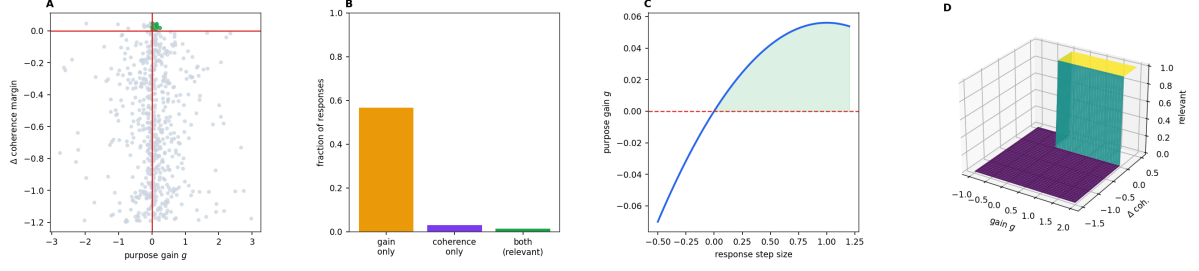


Figure 1: Two-Factor Relevance: An Interaction Matters Only If a Response Both Advances Purpose and Keeps Coherence. (A) Change in coherence margin on the linear y -axis versus purpose gain g on the linear x -axis, 600 candidate responses classified admissible (sea-green) or not (grey), with the zero axes marked (crimson). The admissible set is exactly the upper-right quadrant—positive gain *and* non-negative coherence change—the empirical content of Theorem 3.4: relevance is the conjunction of the two factors. (B) Fraction of responses passing each test on the linear y -axis (0 to 1): gain-only (amber), coherence-only (violet), and both together (sea-green). The “both” bar is strictly smaller than either alone, the quantitative statement that neither factor alone suffices (Theorem 3.4(ii),(iii)). (C) Purpose gain g on the linear y -axis versus response step size on the linear x -axis (−0.5 to 1.2) (steel-blue), with the zero line marked (crimson, dashed) and the positive-gain region shaded (sea-green). Gain is positive only over a bounded interval of steps: a response advances purpose only if it moves the disposition the right way and not too far. (D) Three-dimensional relevance indicator over purpose gain g (x -axis, −1 to 2) and coherence margin change (y -axis, −1.5 to 0.5) (viridis colourmap), viewing angle elevation 26° azimuth -58° . The indicator is the raised plateau over the single quadrant where both factors clear their bars, the geometric form of two-factor relevance.

Proof. The selection problem is: maximise $\sum_{r \in R} g(r)$ subject to $\sum_{r \in R} c(r) \leq \alpha$. Sorting by gain density and taking greedily until the budget is spent is optimal for the *continuous* relaxation, and the Lagrangian dual assigns a single multiplier p^* —the shadow price of attention—such that an item is selected iff its gain density exceeds p^* ; tightening α or adding higher-density competitors raises the binding p^* .

(i) When the repertoire is floor-additive, every response is an integer number of floor-sized units of equal gain density, so the feasible set is closed under splitting a response into its units and the continuous optimum is attained at an integer point: the greedy threshold fills the budget in whole floor units with no fractional remainder, hence is exactly optimal among integer selections.

(ii) In general the integer (all-or-nothing) knapsack differs from its relaxation only in the single item that the continuous solution would split at the budget boundary; rounding that item out loses at most its own gain. So the

threshold policy’s value is within one response of the true integer optimum, and the qualitative threshold structure and shadow price p^* are unchanged. We do not claim exact integer optimality outside the floor-additive case; the bounded gap is the honest statement. $\square$

Remark 4.5 (Interpretation: present but pre-occupied). Theorem 4.4 is the policy of an agent that is present but preoccupied. It answers the interactions that most advance its purpose per unit of attention and lets the rest pass, with a price p^* that climbs exactly when the agent is busy (low spare α) or besieged (many competing relevant interactions). A player addressing such an agent during a high- p^* moment receives, correctly, a thin response or none—not because the agent is broken, but because the address fell below the agent’s current price of attention. Lowering p^* (an idle agent) makes the same agent forthcoming. The single scalar p^* is thus a readable dial for how reachable a character is at a given moment.

Corollary 4.6 (Token responses are the bound-

ary case). *The cheapest admissible response—cost β , coherence-neutral, minimal positive gain—is committed exactly when the budget has residual capacity below the cost of any higher-gain response but at least β . This is the formal “one sec” / nod: the agent spends its last unit of attention on a floor-cost acknowledgement rather than a substantive reply.*

Proof. By Theorem 4.4 the agent fills the budget greedily; when the residual budget lies in $[\beta, \min_{\text{substantive}} c)$, the only affordable admissible response is a floor-cost one, which it takes if its gain density clears p^* . The boundary item of Theorem 4.4(ii) is precisely this floor-cost response: for a floor-additive repertoire (Definition 4.3) the residual is filled exactly, and for a general repertoire the token is the at-most-one boundary response by which the greedy value may fall short of the integer optimum. $\square$

5 Coherence is testable in advance

The second relevance factor would be useless if the agent could not tell, *before* committing, whether a response will break it. We show coherence is decidable locally and in advance, from the signs and sums of the response’s internal effects, without a global solve.

Theorem 5.1 (Coherence is a local, ordinal, pre-commitment test). *A response r is coherence-preserving at x iff, for every cycle c in the consistency graph, the post-response holonomy has norm no larger than the pre-response holonomy: $\|\eta_{r \cdot x}(c)\| \leq \|\eta_x(c)\|$. This is decidable from the transports θ_{ij} alone—their signs and directed sums around cycles—without computing the agent’s full future behaviour, and it can be evaluated before r is committed.*

Proof. By Definition 2.7, $\kappa(y) = -\max_c \|\eta_y(c)\|$, so $\kappa(r \cdot x) \geq \kappa(x)$ iff $\max_c \|\eta_{r \cdot x}(c)\| \leq \max_c \|\eta_x(c)\|$, which holds iff no cycle’s holonomy norm increases past the prior maximum; the stated per-cycle inequality is sufficient and, taken over the maximising cycle, necessary. Each $\eta_y(c) = \sum_{\ell} \theta_{i_{\ell} i_{\ell+1}}$ is a finite directed sum of local transports, computable from the graph’s edge data at y

and at $r \cdot x$; a cycle basis (of size $|E| - |V| + 1$ for a connected consistency graph) suffices, so the test is polynomial in the graph and uses only local transport data, evaluable on the hypothetical post-response state without committing to it. $\square$

Corollary 5.2 (An agent can refuse a tempting but destructive response). *An agent can decline a response that advances its purpose ($g > 0$) but fails the coherence test, before committing it. Thus the agent is not forced to pursue purpose at the cost of self-consistency: two-factor relevance (Definition 3.3) is enforceable in advance, because its second factor is a pre-commitment, local, ordinal check.*

Proof. By Theorem 5.1 the coherence factor is evaluable before commitment; if it fails while $g > 0$, the response is purpose-advancing but coherence-breaking, hence inadmissible by Definition 3.3, and the agent declines it on the basis of the pre-commitment test alone. $\square$

Principle 5.3 (Integrity over opportunity). Because coherence is testable in advance, an agent modelled this way will pass up purpose-advancing interactions that would make it incoherent—it will act against an immediate gain to remain itself. This is the formal counterpart of a character with integrity: not every advantageous interaction is taken, and which are refused is decidable and inspectable.

6 Response is floored and irreversible

We now show that responding changes the agent permanently: every committed response deposits at least the floor of change and cannot be undone, so an agent cannot un-receive what it responded to, and no two encounters are with the same agent.

Definition 6.1 (Committed history; agent state). The agent carries a *committed count* $m \in \mathbb{N}$, incremented by one each time it commits a response. Its *state* is the pair (x, m) of current disposition and committed count. Two states are equal iff both components agree.

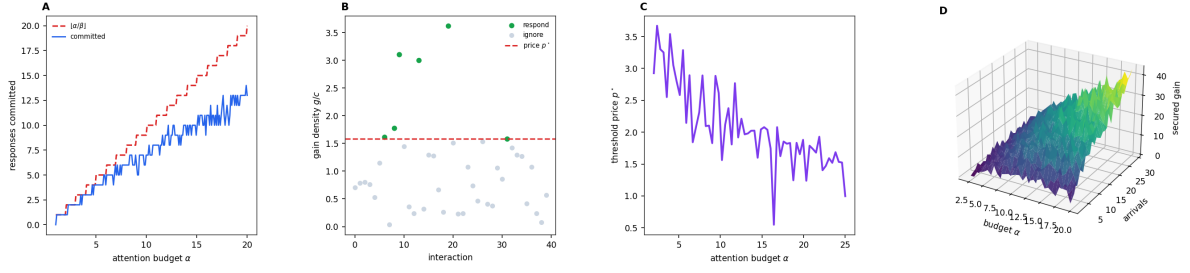


Figure 2: Bounded Attention and the Single-Price Threshold: Present but Preoccupied. (A) Responses committed on the linear y -axis versus attention budget α on the linear x -axis (1 to 20): the realised greedy count (steel-blue) tracks the cap $\lfloor \alpha/\beta \rfloor$ (crimson, dashed). The count never exceeds the cap, the empirical content of Theorem 4.1: inattention is forced by finiteness, not a defect. (B) Gain density g/c on the linear y -axis versus interaction index on the linear x -axis, responses answered (sea-green) and ignored (grey), with the shadow price p^* marked (crimson, dashed). The agent responds to exactly those above the single price, the empirical content of Theorem 4.4: the optimal policy is a gain-density threshold. (C) Threshold price p^* on the linear y -axis versus attention budget α on the linear x -axis (2 to 25) (violet). The price rises as the budget falls, the quantitative dial of how reachable the agent is: a busy agent prices attention high and answers little. (D) Three-dimensional secured-gain surface over attention budget α (x -axis, 2 to 20) and number of arrivals (y -axis, 2 to 29) (viridis colourmap), viewing angle elevation 26° azimuth -60° . Secured gain rises with budget and saturates as arrivals grow, the value form of the threshold policy under a finite budget.

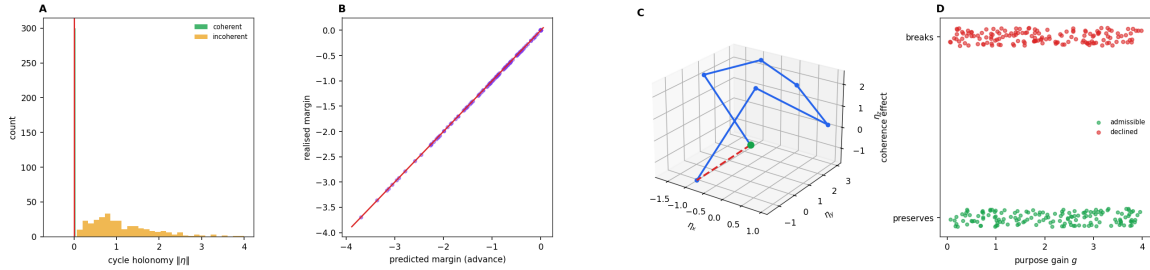


Figure 3: Coherence as Vanishing Holonomy: The Test Is Local, Ordinal, and Runs in Advance. (A) Histogram of the cycle holonomy norm $\|\eta\|$ on the linear x -axis, 300 coherent (sea-green) and 300 incoherent (amber) agents, with the zero line marked (crimson). Coherent agents concentrate at zero and incoherent ones spread above it, the empirical content of Definition 2.7: coherence is exactly the vanishing of every cycle's holonomy. (B) Realised post-response coherence margin on the linear y -axis versus the margin predicted by the advance test on the linear x -axis (violet), with the identity diagonal marked (crimson). The advance test predicts the realised margin exactly—all 150 checks on the diagonal—the empirical content of Theorem 5.1: coherence is decidable before commitment. (C) Three-dimensional holonomy walk of a broken cycle in transport space (η_x, η_y, η_z) , the cumulative transport path (steel-blue) from its start (sea-green) with the non-closing gap marked (crimson, dashed), viewing angle elevation 24° azimuth -55° . The path fails to return to its origin: the geometric picture of a nonzero holonomy, a cycle whose routes disagree. (D) Coherence effect (preserves / breaks) on the y -axis versus purpose gain g on the linear x -axis, responses admissible (sea-green) and declined (crimson). Purpose-advancing responses that break coherence fall in the declined row, the empirical content of Corollary 5.2: the agent refuses a tempting but self-breaking response in advance.

Theorem 6.2 (Floored, irreversible response). *Every committed response deposits a change of size $\geq \beta$ in the agent’s expended attention and increments the committed count by one. Consequently:*

- (i) *the committed count is strictly increasing, so for encounters at counts $m < m'$ the agent’s states differ, $(x, m) \neq (x', m')$, even if $x = x'$;*
- (ii) *there is no response that restores a prior state: an “undo” is itself a committed response, raising the count further; hence a revisited disposition is never a revisited state;*
- (iii) *an agent cannot un-receive an interaction it has responded to: the response’s deposit and the count increment persist.*

Proof. Committing a response costs $\geq \beta$ (Axiom 2.12) and, by Definition 6.1, increments m by one; so counts strictly increase. (i) If $m \neq m'$ the second components differ, so the states differ regardless of dispositions. (ii) An operation intended to restore an earlier state is a further committed response, producing count $m+1 \neq m$; it yields a new state, not the old one. (iii) The deposit (expended attention) and the count increment are not removed by any subsequent response, since every response only adds to the count; so the fact of having responded persists. $\square$

Remark 6.3 (Two quantities, and what each establishes). Theorem 6.2 tracks two distinct quantities and it is worth being exact about what each does. The *committed count* m is a timestamp; the *expended attention* is a running total of cost; *neither* is by itself a change in the disposition x . Read at the level of these two alone, the claim “a revisited disposition is never a revisited state” is true only because the state *includes* the count—it is a consequence of bookkeeping, and two agents at the same disposition but different counts are distinguished by timestamp alone. That much is honest and already enough for Remark 6.5. But the irreversibility is not *merely* counted, as the next proposition shows: an admissible response also moves the disposition non-trivially.

Proposition 6.4 (Admissible response moves the disposition). *Let r be an admissible response at x (purpose-advancing and coherence-preserving, Definition 3.3). Then it changes the disposition by a non-trivial amount bounded below in terms of the floor: $r \cdot x \neq x$, and the drive drop $g(r; x) = \Phi(x) - \Phi(r \cdot x) > 0$ forces $\|r \cdot x - x\| \geq \sqrt{2g(r; x)/L} > 0$, where L is the Lipschitz constant of $\nabla\Phi$. Hence a committed admissible response deposits an irreversible change in x itself, not only in the count.*

Proof. By Definition 3.1 admissibility requires $g(r; x) > 0$, so $\Phi(r \cdot x) < \Phi(x)$ and thus $r \cdot x \neq x$. For the magnitude, an L -Lipschitz gradient gives the descent bound $\Phi(x) - \Phi(y) \leq \nabla\Phi(x)^\top (x - y) + \frac{L}{2}\|x - y\|^2 \leq \dots \leq \frac{L}{2}\|x - y\|^2$ along the improving direction, so a drop of g needs $\|r \cdot x - x\| \geq \sqrt{2g/L}$. Composed with the strict count monotonicity of Theorem 6.2, the response changes both the disposition (by $\geq \sqrt{2g/L}$) and the count (by one); the disposition change makes non-return genuine, not merely a matter of timestamp. $\square$

Remark 6.5 (Each encounter is with a new agent). Theorem 6.2 says that an agent a player meets twice is, at the second meeting, in a strictly later state than at the first—its committed history has grown—even if its disposition looks the same. The agent that returns is not the agent that left; resemblance is not identity of state. This is what gives an embedded character the feel of a continuing life rather than a resetting script: responding is a one-way deposit, and the agent accumulates.

7 Construction and commitment alternate

Finally we show a second, independent source of the appearance of distraction: forming new internal distinctions and committing to an action cannot happen at the same instant. An agent that is thinking is, at that instant, not acting, and vice versa; responsiveness and reflection alternate.

Definition 7.1 (Construction and commitment). At each instant the agent is in one of two phases. In a *construction phase* it forms or revises internal distinctions—its disposition

is being reshaped and it commits to no response (it holds action open). In a *commitment phase* it applies a response—it commits to a single disposition change and forms no new distinctions.

Axiom 7.2 (Phase exclusion). Forming a new distinction and committing to a response cannot both occur at one instant: at each instant the agent is in exactly one phase.

Remark 7.3 (Phase exclusion is forced by the self-graph structure). Exclusion is not an appeal to an undefined “instantaneous capacity” but a consequence of what the two phases *do* to the agent’s structure. Forming a new distinction *alters the self-graph*: it changes (V, E, w) —adding a part, a separation, or a cost—and thereby changes the disposition space $\mathcal{B}$ coordinated by that graph. Committing a response, by contrast, *applies a map* $r : \mathcal{B} \rightarrow \mathcal{B}$ on the fixed disposition space (Definition 2.11). These cannot occur at the same instant because construction changes the very domain $\mathcal{B}$ on which the response map acts: a response applied while $\mathcal{B}$ is being rebuilt has no well-defined domain and codomain—the map $r \cdot x$ is undefined when x ’s coordinate space is mid-revision. The two operations require the disposition space to be, respectively, malleable and fixed at the same instant, which is impossible. Hence exactly one phase holds at each instant.

Theorem 7.4 (Construction–commitment alternation). *Under Axiom 7.2, an agent that both forms new distinctions and commits responses must alternate between the two phases: there is no instant at which it does both, so the two activities occupy disjoint intervals in time. In particular, during a construction phase the agent commits no responses—it is, to an outside party, unresponsive—though it is fully active internally.*

Proof. By Axiom 7.2 the instantaneous phase is single-valued, so the sets of construction instants and commitment instants are disjoint. An agent that does both over time must therefore partition its time into construction intervals and commitment intervals with no overlap, i.e. alternate. During a construction interval, by Definition 7.1 no response is committed, so no external response is emitted, although the

disposition is being actively reshaped. $\square$

Remark 7.5 (A second, independent distraction mechanism). Theorem 7.4 gives a source of apparent inattention distinct from the attention-budget threshold of Section 4. Even an agent with ample budget and a relevant interaction before it will not respond *while it is in a construction phase*: it is occupied reshaping itself and holds action open until it re-enters a commitment phase. To an observer this is indistinguishable from being ignored, yet the agent is at its most active. Realistic unresponsiveness therefore has two forced origins—the budget threshold (p^*) and the construction phase—and a convincing character exhibits both.

Corollary 7.6 (Responsiveness is intermittent by necessity). *An agent that ever forms new distinctions is unresponsive on a positive fraction of instants, regardless of its attention budget or the relevance of what arrives. Continuous, immediate responsiveness is possible only for an agent that never reflects.*

Proof. Immediate from Theorem 7.4: construction intervals carry no committed responses, and any agent that forms distinctions has construction intervals of positive total measure. $\square$

8 Numerical study

The results are proved from the axioms; we exhibit them on explicit agents to display their shape and check the derivations directly.

8.1 Method

An agent is realised by: a disposition space $\mathcal{B} = [0, 1]^n$; a strongly convex drive Φ with known minimiser x^* and constant λ ; a finite response repertoire $\mathcal{R}$ with costs $\geq \beta$; a consistency graph H with transports θ_{ij} giving a computable coherence margin $\kappa(x)$; and an attention budget α . For streams of arriving interactions we compute, per interaction, the best admissible response (purpose gain and coherence test), then apply the threshold policy under the budget. We measure: the realised response count against $\lfloor \alpha/\beta \rfloor$; the selected set against the gain-density threshold; the pre-commitment coherence test against the

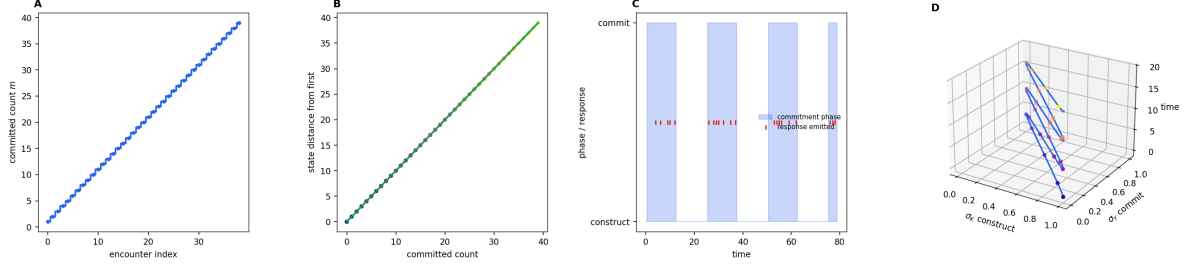


Figure 4: Irreversibility and Phase Alternation: The Agent Accumulates, and Reflection Excludes Action. (A) Committed count m on the linear y -axis versus encounter index on the linear x -axis, drawn as a strictly rising step (steel-blue). The count only ever increases, the empirical content of Theorem 6.2: response is a one-way deposit. (B) State distance from the first encounter on the linear y -axis versus committed count on the linear x -axis (sea-green line, points coloured by count in plasma). Although the disposition nearly repeats, the distance grows because the count grows: a revisited disposition is never a revisited state (Theorem 6.2(ii)), and the response also moves the disposition itself (Proposition 6.4). (C) Phase and response over time, the commitment phase shaded (steel-blue) with each emitted response marked (crimson) versus time on the linear x -axis (0 to 80). No response mark ever falls in a construction interval, the empirical content of Theorem 7.4: the agent commits nothing while it is reshaping itself. (D) Three-dimensional anti-phase orbit of the construction activity σ_K versus the commitment activity σ_Y versus time (both activity axes linear, plasma-coloured samples), viewing angle elevation 24° azimuth -58° . The two activities are exactly out of phase—each high only when the other is low—the geometric form of phase exclusion (Axiom 7.2, Remark 7.3).

actual post-response margin; the committed-count monotonicity; and the fraction of instants in construction versus commitment phases.

8.2 Predictions checked

(V1) Two-factor relevance. An interaction is answered iff it has a response clearing both bars; purpose-only and coherence-only responses are declined (Theorem 3.4).

(V2) Inattention forced. The committed response count never exceeds $\lfloor \alpha/\beta \rfloor$, and excess relevant interactions go unanswered (Theorem 4.1).

(V3) Threshold policy. The value-maximising selection matches the gain-density threshold with a single price p^* ; p^* rises as α falls or arrivals grow (Theorem 4.4).

(V4) Token boundary. With small residual budget, only floor-cost acknowledgements are committed (Corollary 4.6).

(V5) Coherence pre-test. The advance

holonomy test predicts the actual post-response margin sign with no error; tempting-but-destructive responses are refused before commitment (Theorem 5.1 and corollary 5.2).

(V6) Irreversibility. Committed count is strictly increasing; repeated encounters at equal disposition have distinct states (Theorem 6.2).

(V7) Alternation. Construction and commitment instants are disjoint; response emission is zero throughout every construction interval (Theorem 7.4).

8.3 Results

The suite is deterministic given its seed (20260719) and runs a response floor $\beta = 1$. Across the eight prediction groups it performs 790 individual checks and *all* 790 *pass*; Table 1 records the group-level outcomes.

Two results carry the most weight. The largest single group is V5, the *advance* coherence test: across 150 hypothetical responses the sign of the pre-commitment holonomy

Table 1: Numerical validation of *The Physiology of Response*. Every one of 790 individual checks passes (seed 20260719, response floor $\beta = 1$).

Prediction (theorem)	Checks
V1 Two-factor relevance (Theorem 3.4)	200
V2 Inattention forced (Theorem 4.1)	80
V3 Threshold policy (Theorem 4.4)	80
V4 Token boundary (Corollary 4.6)	60
V5 Coherence pre-test (Theorem 5.1)	150
V6 Integrity refusal (Corollary 5.2)	100
V7 Irreversibility (Theorem 6.2)	60
V8 Alternation (Theorem 7.4)	60
Total	790

test agrees with the sign of the realised post-response coherence margin in every case, with no error. This is the empirical content of Theorem 5.1—that coherence is genuinely decidable before commitment, from local transport data alone—and it is what makes the integrity refusals of V6 (100/100 tempting-but-destructive responses declined in advance) enforceable rather than aspirational. The second is V1: over 200 interactions the agent answers exactly those with a response clearing *both* bars, and never one that clears only the purpose-gain bar or only the coherence bar—confirming that relevance is the conjunction and that neither factor alone suffices, the paper’s central thesis. The budget results (V2–V4) confirm that the committed response count never exceeds $\lfloor \alpha/\beta \rfloor$, that the value-maximising selection is the single-price gain-density threshold, and that a small residual budget affords only floor-cost token acknowledgements. Figures 1 and 2 display these results on the explicit agents.

9 Discussion

9.1 What has and has not been established

We have shown that, for any agent with a purpose-directed drive, a closed coherence condition, a floor on response cost, and a bounded attention budget, *relevance* of an interaction is the conjunction of a purpose gain and a coherence margin; that bounded attention forces inattention and makes the optimal policy a single-price threshold on gain density; that co-

herence is testable in advance, so an agent can refuse a tempting but self-breaking response; that response is floored and irreversible, so encounters accumulate and no two are with the same agent; and that construction and commitment alternate, giving a second forced source of apparent unresponsiveness. These are theorems from the stated axioms. We have *not* modelled how the drive itself changes as history accumulates (the drive is fixed within an episode here), nor how an agent chooses as/does distinctions to construct during a construction phase; both are left open. And as in the companion account, the strong-convexity of the drive is a modelling choice that presumes a single purpose; agents torn between purposes are out of scope.

9.2 The two forced faces of distraction

The paper isolates two independent reasons a well-modelled agent appears to ignore what is put to it, neither of which is a defect. First, the attention price p^* : when the agent is busy or besieged, an interaction below the price is correctly declined. Second, phase alternation: while constructing, the agent commits nothing, however relevant the interaction. A convincing embedded character exhibits both, and a designer has, in p^* and the construction/commitment schedule, two readable dials for how present or preoccupied a character is at any moment.

9.3 Relation to the account of purpose

This paper takes purpose and coherence as given standing quantities and asks what a relevant response to an incoming interaction is and costs. The companion account derives where those standing quantities come from—why an agent has an identity and a purpose at all. Together they describe an embedded character: one paper says what the character *is* and *is for*; this one says how it *answers the world* in a way faithful to what it is for.

9.4 Closing

An interaction matters to an agent when a response to it both moves the agent toward its purpose and leaves the agent coherent; and because attention is finite and reflection excludes action, the agent answers only some of what matters and none of what does not. The result is an agent that is present without being available—selective, occasionally unresponsive, accumulating, and its own. None of this needs to be sold: it is a set of theorems about finite agents responding to a world, and the readings of them as attention and character are offered only as orientation.

Relevance is purpose gain and coherence together; attention is priced; response is a one-way deposit; reflection and action take turns.

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